Homework 4

Graded

Student

Colin Skinner

Total Points

79.5 / 80 pts

Question 1

Problem 1

15 / 15 pts

✔ **– 0 pts** Correct

– 3 pts Setup Error

– 3 pts Error in the derivation of EOM

– 7 pts Missing major steps

– 4 pts Incorrect mass ratio

Question 2

Problem 2

15 / 15 pts

2.1 a

10 / 10 pts

✔ **– 0 pts** Correct

– 1 pt Minor/Sign Error

– 2 pts Algebra Error

– 3 pts Setup Error

– 3 pts Incorrect Lagrangian

– 5 pts Incomplete

2.2 b

5 / 5 pts

✔ **– 0 pts** Correct

– 0.5 pts did not set the product of derivatives to zero

– 0.5 pts Minor Error

– 2 pts Incorrect equation from part a

– 3 pts did not make the correct small angle approximation

– 5 pts no attempt

Question 3

Problem 3 — **14.5** / 15 pts

3.1 a — **9.5** / 10 pts

- **– 0 pts** Correct
- ✔ **– 0.5 pts** Minor Error
- **– 2 pts** Incorrect generalized force
- **– 3 pts** Does not solve for generalized force

3.2 b — **5** / 5 pts

- ✔ **– 0 pts** Correct

Question 4

Problem 4 — **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** Does not separate psi from local spherical coordinate system, or incorrect use of angles phi, psi, and theta
- **– 1 pt** Minor error

Question 5

Problem 5 — **20** / 20 pts

- ✔ **– 0 pts** Correct
- **– 10 pts** Did not make Lagrangian formulation.
- **– 5 pts** No comparison of Lagrange to force balance
- **– 20 pts** No attempt

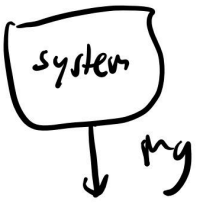
Question assigned to the following page: 1

…ction of the … of the fuels … s expended.

Show that the equation of motion for a rocket projected vertically upward in a uniform gravitational field, neglecting friction, is

$$m\frac{dv}{dt} = -v'\frac{dm}{dt} - mg$$

where m is the mass of the rocket and v' is the velocity of the escaping gases relative to the rocket. Integrate this equation to obtain v as a function of m, assuming a constant time rate of loss of mass. Show, for a rocket starting initially from rest, with v' equal to 2.1 km/s and a mass loss per second equal to 1/60th of the initial mass, that in order to reach the escape velocity the ratio of the weight of the fuel to the weight of the empty rocket must be almost 300!



only external force is gravity

v': exhaust velocity rel. to rocket

$$\vec{v}_e = \vec{v}_r + \vec{v}_{e/r} = \vec{v}_r + \vec{v}'$$

$$\dot{m}_r + \dot{m}_e = 0$$

For +y: $v_e\hat{j} = v_r\hat{j} - v'\hat{j} \rightarrow v' = v_r - v_e$

$$\Sigma F = -mg = \frac{dp_{sys}}{dt} = \frac{dp_r}{dt} + \frac{dp_e}{dt} = \dot{m}_r v_r + m\dot{v}_r + \dot{m}_e v_e + m_e\dot{v}_e$$ (const. v_e)

$$-mg = \dot{m}_r v_r + m\dot{v}_r + \dot{m}_e v_e = \dot{m}_r v_r + m_r\dot{v}_r - \dot{m}_r v_e$$

$$-mg = \dot{m}_r(v_r - v_e) + m_r\dot{v}_r = \dot{m}_r v' + m_r\dot{v}_r = \frac{dm_r}{dt}v' + m_r\frac{dv_r}{dt}$$

($v_r - v_e = v'$)

$$M\frac{dv}{dt} = -v'\frac{dm}{dt} - mg$$

Escape velocity

no gravity — Escape grav. well

$$V_0 + T_0 = V_f + T_f \rightarrow mgh_0 + \frac{1}{2}mv_{esc}^2 = mgh_f + \frac{1}{2}mv_f^2$$

$$g = -G\frac{m_E}{r^2}$$

$$\rightarrow -mG\frac{m_e}{r^2}\cdot r + \frac{1}{2}mv_{esc}^2 = 0 \qquad v_{esc}^2 = \frac{2Gm_e}{r} \qquad v_{esc} = \sqrt{\frac{2Gm_e}{r}}$$

$$M\frac{dv}{dt} = -v'\frac{dm}{dt} - mg$$

$$\int_0^v dv = \int_{m_0}^{m} -\frac{v'}{m}dm - \int_0^{t_b}\frac{mg}{m}dt$$

$$v = -v'\ln\frac{m}{m_0} - gt = -v'\ln\frac{m}{m_0} - g\frac{m}{\dot{m}}$$

$$v = v'\ln\left(\frac{m_0}{m}\right) - gt$$

Question assigned to the following page: 1

$v \ln\left(\frac{m_0}{m_f}\right) \; \ldots \; r_e$

$t_b = 60\,s$

$m_0 = m_s + m_f$

$m_f = m_s$

$$v' \ln\left(\frac{m_s + m_f}{m_s}\right) = \sqrt{\frac{2GM_e}{r_e}} + g t_b$$

$$\ln\left(\frac{m_f}{m_s} + 1\right) = \frac{1}{v_1}\left(\sqrt{\frac{2GM_e}{r_e}} + g t_b\right)$$

$$\frac{m_f}{m_s} + 1 = e^{\frac{1}{v_1}\left(\sqrt{\frac{2GM_e}{r_e}} + g t_b\right)}$$

$$\frac{m_f}{m_s} = e^{\frac{1}{v_1}\left(\sqrt{\frac{2GM_e}{r_e}} + g t_b\right)} - 1$$

$$\frac{m_f}{m_s} = 270.8$$

$t_b = 60\,s$

$G = 6.67E{-}11 \; \frac{m^3}{kg\,s^2}$

$M_e = 5.972E24 \; kg$

$r_e = 6371E3 \; m$

$v' = 2.1E3 \; \frac{m}{s}$

$g = 9.8 \; \frac{m}{s^2}$

Question assigned to the following page: 2.1

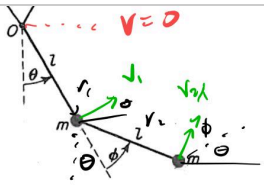


Fig. P6-4

A double pendulum consists of two massless rods of length l and two particles of mass m which can move in a given vertical plane, as shown.

a. Assuming frictionless joints, and using θ and φ as coordinates, obtain the differential equations of motion.

b. What are the linearized equations for small θ and φ?

$$\vec{r}_1 = l\sin\theta\,\hat{x} - l\cos\theta\,\hat{y}$$

$$\dot{\vec{r}}_1 = l\dot{\theta}\cos\theta\,\hat{x} + l\dot{\theta}\sin\theta\,\hat{y}$$

$$\vec{r}_2 = \left(l\sin\theta + l\sin(\theta+\phi)\right)\hat{x} - \left(l\cos\theta + l\cos(\theta+\phi)\right)\hat{y}$$

$$\dot{\vec{r}}_2 = \left(l\cos\theta\,\dot{\theta} + l(\dot{\theta}+\dot{\phi})\cos(\theta+\phi)\right)\hat{x} + \left(l\dot{\theta}\sin\theta + l(\dot{\theta}+\dot{\phi})\sin(\theta+\phi)\right)\hat{y}$$

$$v_1^2 = l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta = l^2\dot{\theta}^2$$

$$v_2^2 = \left(l\dot{\theta}\cos\theta + l(\dot{\theta}+\dot{\phi})\cos(\theta+\phi)\right)^2 + \left(l\dot{\theta}\sin\theta + l(\dot{\theta}+\dot{\phi})\sin(\theta+\phi)\right)^2$$

$$= l^2\dot{\theta}^2\cos^2\theta + 2l^2\dot{\theta}(\dot{\theta}+\dot{\phi})\cos(\theta+\phi)\cos\theta + l^2(\dot{\theta}+\dot{\phi})^2\cos^2(\theta+\phi)$$

$$+ l^2\dot{\theta}^2\sin^2\theta + 2l^2\dot{\theta}(\dot{\theta}+\dot{\phi})\sin(\theta+\phi)\sin\theta + l^2(\dot{\theta}+\dot{\phi})^2\sin^2(\theta+\phi)$$

$$= l^2\dot{\theta}^2 + l^2(\dot{\theta}+\dot{\phi})^2 + 2l^2\dot{\theta}(\dot{\theta}+\dot{\phi})\left(\cos(\theta+\phi)\cos\theta + \sin(\theta+\phi)\sin\theta\right)$$

$\cos[(\theta+\phi)-\theta] = \cos\phi$

$$= l^2\dot{\theta}^2 + l^2(\dot{\theta}+\dot{\phi})^2 + 2l^2\dot{\theta}(\dot{\theta}+\dot{\phi})\cos\phi$$

$$T = \frac{1}{2}ml^2\left[2\dot{\theta}^2 + (\dot{\theta}+\dot{\phi})^2 + 2\dot{\theta}(\dot{\theta}+\dot{\phi})\cos\phi\right] \qquad \mathcal{L} = T - V$$

$$V_2 = mg(-l\cos\theta) + mg\left(-l\cos(\theta+\phi)\right) = -mgl\left(2\cos\theta + \cos(\theta+\phi)\right)$$

$$-\frac{\partial \mathcal{L}}{\partial\theta} = mgl\left(2\sin\theta + \sin(\theta+\phi)\right)$$

$$\frac{\partial \mathcal{L}}{\partial\dot{\theta}} = \frac{1}{2}ml^2\left[2l\dot{\theta} + 2(\dot{\theta}+\dot{\phi}) + 2(\dot{\theta}+\dot{\phi})\cos\phi + 2\dot{\theta}\cos\phi\right]$$

$$= ml^2\left[3\dot{\theta} + \ddot{\phi} + 2\dot{\theta}\cos\phi + \dot{\phi}\cos\phi\right] = ml^2\left[(3+2\cos\phi)\ddot{\theta} + (1+\cos\phi)\ddot{\phi}\right]$$

Questions assigned to the following page: 2.1 and 2.2

$$\frac{d}{dt}\frac{\partial}{\partial \dot{\theta}} - ml^2\left[-2\dot{\theta}\dot{\phi}\sin\phi + (3+2\cos\phi)\ddot{\theta} + (-\dot{\phi}\cos\phi)\dot{\phi} + (1+\cos\phi)\ddot{\phi}\right]$$

$$ml^2\left[-2\dot{\theta}\dot{\phi}\sin\phi + (3+2\cos\phi)\ddot{\theta} - \dot{\phi}^2\sin\phi + (1+\cos\phi)\ddot{\phi}\right] + \frac{mgl}{l}\left(2\sin\theta + \sin(\theta+\phi)\right) = 0$$

$$(3+2\cos\phi)\ddot{\theta} + (1+\cos\phi)\ddot{\phi} - 2\dot{\theta}\dot{\phi}\sin\phi - \dot{\phi}^2\sin\phi + \frac{1}{l}\left(2\sin\theta + \sin(\theta+\phi)\right) = 0$$

$$-\frac{\partial L}{\partial \phi} = +\frac{1}{2}ml^2\left(2\dot{\theta}(\dot{\theta}+\dot{\phi})(-\sin\phi)\right) - mgl\left(-\sin(\theta+\phi)\right)$$

$$= ml^2\ddot{\theta}(\dot{\theta}+\dot{\phi})\sin\phi + mgl\sin(\theta+\phi)$$

$$\frac{\partial L}{\partial \dot{\phi}} = \frac{1}{2}ml^2\left[2(\dot{\theta}+\dot{\phi}) + 2\dot{\theta}\cos\phi\right] = ml^2\left((1+\cos\phi)\dot{\theta} + \dot{\phi}\right)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\phi}} = ml^2\left[(-\sin\phi\dot{\phi})\dot{\theta} + (1+\cos\phi)\ddot{\theta} + \ddot{\phi}\right]$$

$$= ml^2\left[\ddot{\phi} + (1+\cos\phi)\ddot{\theta} - \dot{\phi}\dot{\theta}\sin\phi\right]$$

$$ml^2\left[\ddot{\phi} + (1+\cos\phi)\ddot{\theta} - \dot{\phi}\dot{\theta}\sin\phi\right] + ml^2\ddot{\theta}(\dot{\theta}+\dot{\phi})\sin\phi + \frac{mgl}{l}\sin(\theta+\phi) = 0$$

$$\ddot{\phi} + (1+\cos\phi)\ddot{\theta} - \dot{\phi}\dot{\theta}\sin\phi + \dot{\theta}^2\sin\phi + \dot{\theta}\dot{\phi}\sin\phi + \frac{g}{l}\sin(\theta+\phi) = 0$$

$$\ddot{\phi} + (1+\cos\phi)\ddot{\theta} + \dot{\theta}^2\sin\phi + \frac{1}{l}\sin(\theta+\phi) = 0$$

Small angle

$$(3+2\cos\phi)\ddot{\theta} + (1+\cos\phi)\ddot{\phi} - 2\dot{\theta}\dot{\phi}\sin\phi - \dot{\phi}^2\sin\phi + \frac{1}{l}\left(2\sin\theta + \sin(\theta+\phi)\right) = 0$$

$$5\ddot{\theta} + 2\ddot{\phi} - 2\phi\dot{\phi}\dot{\theta} - \dot{\phi}^2\phi + \frac{g}{l}(2\theta + \theta + \phi) = 0$$

-0 -0

Question assigned to the following page: 2.2

$$5\ddot{\theta} + 2\ddot{\phi} + \frac{g}{\ell}(3\theta + \phi) = 0$$

$$\ddot{\phi} + (1+\cos\phi)\ddot{\theta} + 2\dot{\theta}\sin\phi + \frac{1}{\ell}\sin(\theta+\phi) = 0$$

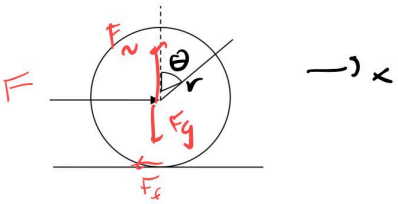
$$\ddot{\phi} + 2\ddot{\theta} + \frac{g}{\ell}(\theta+\phi) = 0$$

Questions assigned to the following page: 3.1 and 3.2

direction parallel to the plane of the disk and parallel to the ground.

a. Derive Lagrange's equations and find the generalized force.

b. Discuss the motion if the force is not applied parallel to the plane of the disk (i.e. now with a component out of/into the page).



Start w/ x and θ

$\dot{x} = r\dot{\theta}$ ← constraint

$Q_x = F_x \cdot \frac{\partial \vec{r}}{\partial x} + F_f \cdot \frac{\partial \vec{r}}{\partial x} + F_g \cdot \frac{\partial \vec{r}}{\partial x} + F_N \cdot \frac{\partial \vec{r}}{\partial x} = F_x - F_f$

$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\left(\frac{1}{2}Mr^2\right)\left(\frac{\dot{x}}{r}\right)^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{4}m\dot{x}^2$

$= \frac{3}{4}m\dot{x}^2 \qquad \mathcal{L} = \frac{3}{4}m\dot{x}^2 \qquad \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{3}{2}m\dot{x} \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{3}{2}m\ddot{x}$

$\frac{\partial \mathcal{L}}{\partial x} = 0$

$$\frac{3}{2}m\ddot{x} = F - F_f$$

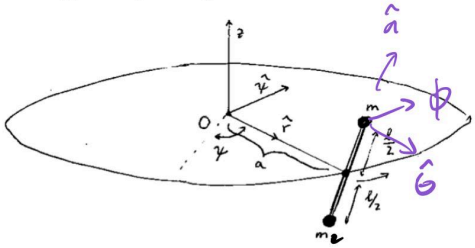
If the force has components in/out of the page, we would need to add another generalized coordinate. In both the x and (new) z coordinates, F and F_f would have components.

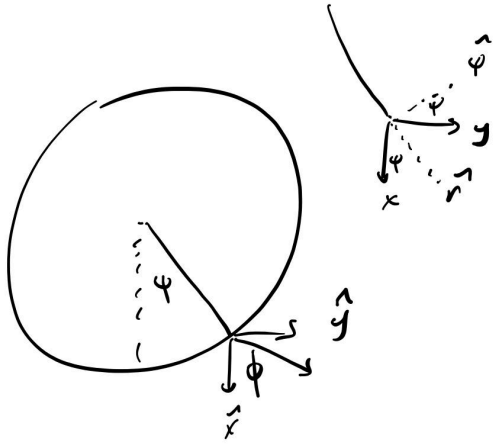
Question assigned to the following page: 4

of mass , are joined by a

move on a circle of radius.

Express the kinetic energy of the system in generalized coordinates.





(Hint: In addition to the cylindrical coordinate system shown, define an *x*-*y*-*z* coordinate system that has a fixed orientation in space but *translates with the rod center of mass*. For each particle, use a spherical coordinate system with angles defined relative to this frame)

$\vec{r} = a\hat{r}$

$\vec{v} = a\dot{\psi}\hat{\psi}$ $\rightarrow$ $T_{com} = \frac{1}{2}(2m)\left(a^2\dot{\psi}^2\right) = ma^2\dot{\psi}^2$

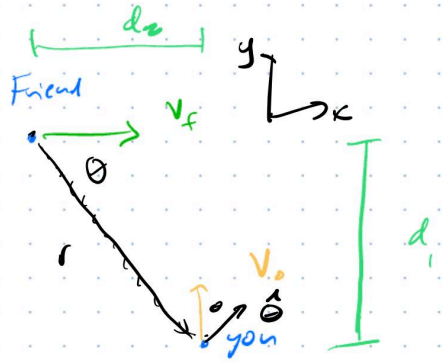
$\vec{v}_{sph} = \left(\frac{\dot{\ell}}{2}\right)\hat{n} + \frac{\ell}{2}\dot{\theta}\hat{\theta} + \frac{\ell}{2}\dot{\phi}\sin\theta\,\hat{\phi} = \frac{\ell}{2}\left(\dot{\theta}\hat{\theta} + \dot{\phi}\sin\theta\,\hat{\phi}\right)$

$T_{sph} = 2\left[\frac{1}{2}m\frac{\ell^2}{4}\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right)\right] = \frac{m\ell^2}{4}\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right)$

$$T = m\left[a^2\dot{\psi}^2 + \frac{\ell^2}{4}\left(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta\right)\right]$$

Question assigned to the following page: 5

problem that you chose and began
ide a short description of what you
are trying to solve. For this week, attempt a solution using Lagrange's approach. Define your generalized coordinates and write the Lagrangian. Use the Lagrangian to solve your EOM. Compare this approach to force balance and discuss. How does your answer this week compare to your answer using force balance? If you are still unable to obtain a solution this week, discuss how you might simplify the problem.



A friend and I are biking past each other on campus. He is traveling at the speed v_f = 4 m/s and I am traveling at the speed 7 m/s. I am at a perpendicular distance of 14 m from his path of travel, and he is 18 meters from mine. At what point are we the closest, and at what angle should my friend wave to me clockwise from his direction of travel?

a) Find equation of motion of you relative to friend

general coordinates: r and θ challenge

coords $\quad \hat{x} = \hat{r}\cos\theta + \hat{\theta}\sin\theta \qquad \hat{y} = -\hat{r}\sin\theta + \hat{\theta}\cos\theta$

$\vec{r} = r\hat{r} \qquad \dot{\vec{r}} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} = \vec{v} \rightarrow v^2 = \dot{r}^2 + r^2\dot{\theta}^2$

kinetic $\quad T = \frac{1}{2}m_o v^2 = \frac{1}{2}m_o(\dot{r}^2 + r^2\dot{\theta}^2) \qquad L = T - V = T = \frac{1}{2}m_o(\dot{r}^2 + r^2\dot{\theta}^2)$

$\frac{\partial L}{\partial r} = \frac{1}{2}m_o(2r\dot{\theta}^2) = m_o r\dot{\theta}^2$

$\frac{\partial L}{\partial \dot{r}} = \frac{1}{2}m_o(2\dot{r}) = m_o\dot{r} \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = m_o\ddot{r}$

$m_o\ddot{r} - m_o r\dot{\theta}^2 = 0$

$\ddot{r} - r\dot{\theta}^2 = 0$

shows acceleration in r direction is 0

Question assigned to the following page: 5

$\frac{\partial L}{\partial \theta} = 0 \quad \frac{\partial L}{\partial \dot{\theta}} = \frac{1}{2} m_0 r^2 (2\dot{\theta}) = m_0 r^2 \dot{\theta} \quad \frac{M^2}{s}$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = m_0 (2 r \dot{r}) \dot{\theta} + m_0 r^2 \ddot{\theta}$

$2 m_0 r \dot{r} \dot{\theta} + m_0 r^2 \ddot{\theta} = 0$

$2\dot{r}\dot{\theta} + r\ddot{\theta} = 0$ show acceleration in θ direction is 0

This didn't really help but it did help me and it will take a while to solve the correct way. Especially with friction

Precias attempt

$\vec{r}_f = -d_f \hat{x} \quad \vec{v}_f = v_f \hat{x} \quad \vec{r}_0 = -d_0 \hat{y} \quad \vec{v}_0 = v_0 \hat{y}$

$\vec{r}_{0/f} = \vec{r}_0 - \vec{r}_f = -d_0 \hat{y} + d_f \hat{x}$

$Q_r' = F_x \cdot \frac{\partial \hat{x}}{\partial r} + F_y \frac{\partial \hat{y}}{\partial r} \qquad Q_\theta' = F_x \frac{\partial \hat{x}}{\partial \theta} + F_y \frac{\partial \hat{y}}{\partial \theta}$

$= -F_{f,f} \cos\theta - F_{f,0}(-\sin\theta) \qquad = -F_{f,f}(-r\sin\theta) - F_{f,0}(r\sin\theta)$

$Q_r' = -F_{f,f}\cos\theta + F_{f,0}\sin\theta \qquad Q_\theta' = F_{f,f} r\sin\theta - F_{f,0} r\sin\theta$

Question assigned to the following page: 5

$\vec{v}_o = \vec{v}_f + \vec{v}_{o/f}$

$\vec{v}_{o/f} = \vec{v}_o - \vec{v}_f = v_o \hat{y} - v_f \hat{x}$

$V = 0$

$T = \frac{1}{2} m v_{o/f}^2 = \frac{1}{2} m \dot{r}^2$ $\quad L = T - V = \frac{1}{2} m \dot{r}^2$

$L = T - V = \frac{1}{2} m (v_o^2 + v_f^2)$

$\frac{\partial L}{\partial r} = 0 \quad \frac{\partial L}{\partial \dot{r}} = m\dot{r} \quad \frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = m\ddot{r}$

$\frac{\partial L}{\partial \theta} = 0 \quad \frac{\partial L}{\partial \dot{\theta}} = 0$

$\hat{r} = \cos\theta \hat{x} - \sin\theta \hat{y}$

$\theta = \tan\left(\frac{d_o}{d_f}\right)$

$Q_j = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j}$

$\vec{\rho}_1 = a\hat{r} + \frac{\ell}{2}\hat{a}$

generalized coordinates: ψ, ϕ, θ

$\hat{x} = \hat{r}\cos\psi - \hat{\psi}\sin\psi$

$\hat{y} = \hat{r}\sin\psi + \hat{\psi}\cos\psi$

$\hat{a} = \sin\theta\cos\phi\hat{x} + \sin\theta\sin\phi\hat{y} + \cos\hat{z}$